THE MASTERMIND COACHING CLASSES

Revision Notes

Class 7 · Science

NCERT Curiosity — Chapters 1 to 6 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Exploring Substances: Acidic, Basic, and Neutral	Indicators, neutralisation, daily-life uses
2	Electricity: Circuits and Their Components	Cell, lamp, switch, circuit, conductors, insulators
3	The World of Metals and Non-metals	Properties, rusting, oxides, uses of elements
4	Changes Around Us: Physical and Chemical	Physical vs chemical change, combustion, weathering
5	Adolescence: A Stage of Growth and Change	Puberty, menstruation, hormones, healthy habits
6	Heat Transfer in Nature	Conduction, convection, radiation, breezes, water cycle

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CHAPTER 01

Exploring Substances: Acidic, Basic, and Neutral

This chapter begins at a science fair where a plain sheet of paper reveals a hidden message when sprayed with a liquid. To explain the trick, it sorts everyday substances into acidic, basic and neutral groups, shows how natural indicators (litmus, red rose, turmeric, olfactory indicators) tell them apart, and then studies what happens when an acid meets a base.

Big Questions

- How can we tell whether a substance is acidic, basic or neutral without tasting it?
- What are natural indicators and how are they made?
- What happens when an acid and a base are mixed?
- Where do we use neutralisation in everyday life?

2.1 Nature — Our Science Laboratory

The siblings Ashwin and Keerthi find that sprayed liquid makes writing appear on blank paper. To understand this they test many household samples: lemon juice, soap solution, amla juice, tamarind water, vinegar, baking soda solution, lime water, tap water, washing powder solution, sugar solution and salt solution.

▸ 2.1.1 Litmus as an indicator (Activity 2.1)

Step 1 — Prepare the samples
Cut blue litmus paper into small pieces and place them on a clean, dry white tile. Do the same with red litmus paper.

Step 2 — Test
Put one drop of each sample on separate pieces of blue and on separate pieces of red litmus paper using a dropper.

Step 3 — Record and group
Note the colour changes in a table, then sort the samples into three groups A, B and C.

Group	What happens	Samples	Nature
Group A	Blue litmus turns **red**	Lemon juice, amla juice, tamarind water, vinegar	**Acidic**
Group B	Red litmus turns **blue**	Soap solution, baking soda solution, lime water, washing powder solution	**Basic**
Group C	No change in either litmus	Tap water, sugar solution, salt solution	**Neutral** (neither acidic nor basic)

DEFINITION

Litmus is a natural substance obtained from **lichens**. It is sold as a solution and as blue and red **litmus paper**. Because it shows different colours in acidic and basic solutions, it is an **acid-base indicator**. Synthetic indicators are made in laboratories (studied in higher classes).

> **NOTE**
> **Lime water** (a solution of calcium hydroxide) is not the same as the fruit lime. It is prepared by mixing lime (*chuna*, calcium oxide) in water, leaving it undisturbed for about an hour and filtering the clear liquid.

Lichens are formed by an association of two living organisms, a **fungus and an alga**. They grow on rocks and trees in regions with plenty of rain and clean air.

▸ Taste and touch as clues (with caution)

- Substances in Group A taste **sour**: sour-tasting substances tend to contain **acids**.
- Bases such as baking soda solution feel **soapy or slippery** between the fingers and generally taste **bitter**.
- But not everything bitter is a base: bitter gourd (*karela*) is bitter yet not basic.
- **Never taste unknown substances** and never taste anything unless the teacher asks you to.

Edible substance	Common acid present
Lemon, orange (citrus fruits)	Citric acid
Curd	Lactic acid
Tamarind	Tartaric acid
Vinegar	Acetic acid
Amla, citrus fruits	Ascorbic acid (Vitamin C)
Spinach, tomato	Oxalic acid

> **REMEMBER**
> Acids turn **blue litmus red**; bases turn **red litmus blue**. A strong exam habit: write the colour change along with the conclusion.

▸ 2.1.2 Red rose as an indicator (Activities 2.3 and 2.4)

> **Prepare the extract**
> Collect fallen red rose petals (do not pluck flowers), wash and crush them with a mortar and pestle, place in a tumbler, add hot water to cover them (under adult supervision), cover and wait 5–10 minutes until the water is coloured, then filter. The **filtrate** is the rose extract.

> **Test**
> Put 10–20 drops of extract in test tubes and add 20–30 drops of each sample with a dropper. Record the colour.

Sample type	Colour of red rose extract	Matches litmus group
Acidic (e.g. lemon juice)	Shade of **red**	Group A
Basic (e.g. soap solution)	Shade of **green**	Group B
Neutral (e.g. salt solution)	No change	Group C

So red rose extract is another acid-base indicator. Beetroot, **purple cabbage**, turmeric, Indian blackberry (**jamun**) and red **hibiscus** (gudhal) flowers can be used in the same way.

> **NOTE**
> **Hydrangea** flowers (grown in cooler Himalayan and north-eastern regions) show the soil's nature: they are **blue in acidic soil** and **pink or red in basic soil**.

▸ 2.1.3 Turmeric as an indicator (Activity 2.5)

- Make a paste of turmeric (*haldi*) with a little water, dip a filter paper in it until it turns yellow, dry it and cut it into strips: this is **turmeric paper**.
- Turmeric paper turns **red** with **basic** substances (soap, baking soda, lime water).
- It shows **no change** with acidic and neutral substances, so it **cannot distinguish between acidic and neutral** substances; it only tests for bases.
- This is why a curry stain on a white shirt turns red when soap (a base) is rubbed on it.
- Ashwin's greeting card: he wrote on dried turmeric paper with a basic solution, such as soap solution, and the message appeared in red.

Turmeric is called a 'golden spice'. It belongs to the ginger family, is widely grown in India, and is used in Ayurveda and traditional home remedies.

▸ Olfactory indicators (Activity 2.6)

- Some substances change their **odour** in acidic or basic medium; these are **olfactory indicators**.
- Finely chopped onions are kept overnight in a closed container with strips of cotton cloth or filter paper, so that the strips pick up the onion smell.
- When tamarind water (acid) or baking soda solution (base) is dropped on a strip, the onion smell **disappears or changes** in basic medium, while it stays in acidic medium (the smell is lost with a base).

Indicator	In acidic solution	In basic solution	In neutral solution
Blue litmus	Turns red	No change (stays blue)	No change
Red litmus	No change (stays red)	Turns blue	No change
Red rose extract	Red	Green	No change
Turmeric paper	No change (yellow)	Turns red	No change
Onion (olfactory)	Odour remains	Odour lost	Odour remains

> **NOTE**
> **Know a scientist — Acharya Prafulla Chandra Ray (P. C. Ray):** called the 'Father of Modern Indian Chemistry'; earned a doctorate in the UK, returned to advance research in India, set up India's **first pharmaceutical company in 1901**, wrote on the history of chemistry in India and supported teaching in the **mother tongue**.

2.2 What Happens When Acidic Substances Mix with Basic Substances?

Activity 2.7 — set-up
Put one drop of lemon juice in a test tube, add about twenty drops of water, then a drop of blue litmus solution. The solution turns **red** (acidic).

Add the base
Add lime water drop by drop and swirl. After a stage the colour changes from **red to blue**: the solution is no longer acidic, because lime water has **neutralised** the acid.

Reverse it
Add one more drop of lemon juice: the colour goes back to red because the acid is now in excess.

DEFINITION

When an acid solution is mixed with a base solution in the right quantity, the result is neither acidic nor basic. This reaction is **neutralisation**. It forms a **salt and water** and **releases heat**.

FORMULA

Acid + Base → Salt + Water + Heat

2.3 Neutralisation in Daily Life

Situation	Problem	Remedy and reason
Red ant bite	Ant injects **formic acid** into the skin, giving redness and stinging pain	Rub **moist baking soda** (a base), which neutralises the acid
Acidic soil (often from excess chemical fertilisers)	Plants do not grow well	Add **lime** (a base) to the soil
Basic soil	Poor plant growth	Add organic matter such as **manure or composted leaves**, which release acids
Acidic factory waste in a lake	Fish population declines	Neutralise the waste with a **basic substance** before releasing it

Sometimes soil is neutral but plants are still unhealthy: then the cause may be a **deficiency of nutrients**, not acidity or basicity.

Wrapping up the science-fair puzzle

The hidden message trick can be done with a **turmeric solution** in the spray bottle and a **basic solution such as soap solution** used as the invisible ink for writing. The soap turns the yellow turmeric red, so the words appear.

Worked answers to the end-of-chapter questions

Likely question	Model answer
Excess of which solution would reverse red litmus turning blue?	Vinegar (an acid) neutralises the base that made the litmus blue.
A turns red litmus blue, B turns turmeric red, C turns rose extract green. Nature?	All three are **basic**.
Red litmus unchanged, blue litmus turns red, turmeric unchanged. Nature of the liquid?	**Acidic**: acids turn blue litmus red and do not change red litmus or turmeric.
Which indicator can a blindfolded person use?	An **olfactory indicator** such as onion: it depends on smell, not sight.
Grape juice and red rose extract gave a red tint. What happens when baking soda is added?	The base turns the mixture green/changes it from red, showing the juice was acidic and is neutralised.

Likely question	Model answer
How would you reveal a message written with orange juice?	Orange juice is acidic; use a **blue litmus solution or red rose extract** (turns red) so that the writing shows.
Identify vinegar, baking soda solution and sugar solution using only turmeric paper.	Baking soda turns it red. Vinegar and sugar do not change it; distinguish them by **taste or smell** (sour smell of vinegar) because turmeric cannot separate acidic from neutral.
Rose extract turns liquid X green. What is X? What happens with excess amla juice?	X is **basic**; excess amla juice (acid) neutralises it and then turns the extract red.
Why do vinegar and eggshell or marble bubble, but soap solution does not?	The acid reacts with the carbonate in eggshell or marble releasing a gas (carbon dioxide); a base does not.
What do we mean by indicator? Give two examples.	A substance that shows different colours in acidic and basic solutions, e.g. litmus, turmeric or red rose extract.

Projects suggested in the chapter

- Make **rangoli** using acidic or basic substances and natural indicators.
- Test rain, tap and river water for acidic or basic nature.
- Collect soil of your area and test it as acidic, basic or neutral.

★ KEY TAKEAWAYS

- Substances are classified as **acidic, basic or neutral**.
- Blue litmus turns **red** in acids; red litmus turns **blue** in bases; neutrals change neither.
- **Red rose extract**: red in acids, green in bases. **Turmeric paper**: red in bases only.
- **Olfactory indicators** (such as onion) change odour in acidic or basic medium.
- **Acid + Base → Salt + Water + Heat** is neutralisation.
- Neutralisation explains remedies for ant bites, acidic or basic soil and factory waste.
- Never taste unknown substances.

CHAPTER 02

Electricity: Circuits and Their Components

The chapter starts with a class trip to the Bhakra Nangal Dam and a list of the many uses of electricity. It then uses an ordinary torchlight to build the basic ideas: the electric cell and battery, incandescent and LED lamps, a complete electrical circuit, switches, circuit diagrams, and the difference between conductors and insulators.

Big Questions

- Why does a torch lamp glow only in one position of its switch?
- What is a complete electrical circuit?
- Why must an LED be connected a particular way?
- Why are wires made of metal and covered with plastic or rubber?

3.0 Uses of electricity

Use	Examples
Cooking	Electric kettle, mixer-grinder, toaster, oven, microwave
Lighting	Homes, offices, streets, markets, factories
Entertainment	Television, radio
Transportation	Train, bus, car, scooter, lift, escalator
Communication	Mobile phone, Internet
Heating and cooling	Fan, room heater, immersion rod, geyser, refrigerator, air conditioner
Others	Water pump, crane, computer

Electricity is generated by windmills (wind energy), solar panels (Sun's energy), falling water and by burning natural gas or coal. It reaches homes and factories through wires. To study it safely we use a **portable source** such as a cell in a torch.

IMPORTANT

Safety: electricity is dangerous. Never experiment with mains supply or portable generators at home or school. Use only **cells or batteries** (as in torches, wall clocks, radios, remotes).

3.1 A Torchlight

A torch has a **lamp**, a **switch** and two or more **electric cells** inside. In one position of the switch the lamp glows, in the other it does not. To understand why, we study each component.

3.2 A Simple Electrical Circuit

▸ 3.2.1 Electric cell

- An electric cell has two **terminals**: positive (+ve) and negative (–ve).

- The small **metal cap** is the **positive** terminal; the flat **metal disc** is the **negative** terminal.
- A cell is a **portable source of electrical energy**.

▸ 3.2.2 Battery

- In a torch with two cells, the lamp glows only when the cells are placed in the correct order: the **positive terminal of one cell touches the negative terminal of the next**.
- Two or more cells connected like this form a **battery**.
- More cells supply energy for a **longer time** and/or **more energy**.
- In everyday speech even a single cell (e.g. in a mobile phone) is called a battery.
- In devices with cells side by side, a thick wire or metal strip joins the + of one cell to the – of the next; the '+' and '–' signs are printed inside the compartment.

▸ 3.2.3 Electric lamp

Feature	Incandescent lamp	LED lamp
Light-producing part	Thin wire called **filament**, which gets hot and glows	No filament; a light emitting diode
Terminals	Two terminals: metal tip at centre and metal case; they do not touch each other	Two terminals: **longer wire = positive**, **shorter wire = negative**
Which way to connect?	**Either way**; polarity does not matter	**Only one way**; current passes in one direction only
Where used	Old torches and bulbs	Most modern torches; may use one or more LEDs

- In an incandescent lamp the filament is attached to **two thicker supporting wires**: one goes to the metal case at the base and the other to the metal tip at the centre.
- A lamp is said to be **fused** when its filament breaks; a broken filament stops current, so the lamp will not glow even when connected to a cell.

▸ 3.2.4 Making a lamp glow (Activity 3.6)

Step 1 — Gather
Cell, torch lamp, cell holder, lamp holder and four lengths of wire.

Step 2 — Strip wires
Remove about 1 cm of plastic covering from both ends of each wire to expose the metal.

Step 3 — Fit the cell
Fix wires to the cell holder and put the cell with its negative terminal towards the spring side. Without a holder, use electrical tape.

Step 4 — Fit the lamp
Attach two wires to the lamp holder screws (or tape them to the lamp ends).

Step 5 — Predict, then observe
In the book's six arrangements the lamp glows only in **arrangements 1 and 6**, where each lamp terminal is connected to a different terminal of the cell.

▸ 3.2.5 An electrical circuit

> **DEFINITION**
> An **electrical circuit** is a complete path for electric current formed when one terminal of the lamp is joined to one terminal of the cell and the other lamp terminal to the other cell terminal. The lamp glows **only when current passes** through the circuit.

- By convention, the direction of current is **from the positive to the negative terminal** of the cell.
- In an incandescent lamp the current passes through the **filament**, which makes it glow.

▸ Making an LED glow (Activity 3.7)

- Use two cells in a holder (negative terminal towards the spring) and connect the **positive of the battery to the longer wire** of the LED and the negative to the shorter wire. The LED glows.
- Interchange the connections: the LED **does not glow**, because current passes through it in one direction only.
- The positive terminal of a battery made of two cells is the holder terminal connected to the + terminal of one cell; the other is connected to the – of the other cell.

▸ 3.2.6 Electric switch

> **Make a switch (Activity 3.8)**
> Fix a drawing pin through the ring of a safety pin on a cardboard so it can rotate; fix a second drawing pin so the free end of the safety pin can touch it; connect a wire to each pin.

> **Test it (Activity 3.9)**
> Connect cell, lamp and switch. With the pin not touching the second drawing pin the lamp is **off**; when it touches both pins the lamp glows.

Switch position	Circuit state	Current	Lamp
ON	**Closed** (complete)	Flows from + to – of the cell	Glows
OFF	**Open** (air gap)	Does not flow	Does not glow

A **switch** is a simple device that completes or breaks a circuit. It can be placed **anywhere** in the circuit. House switches work the same way though they look different.

3.3 Circuit Diagrams

Component	How its symbol looks / what it means
Electric cell	Two parallel lines: **long line = positive**, **short thick line = negative**
Battery	Two or more cell symbols joined
Electric lamp	A circle with a cross inside
LED	A triangle pointing in the direction current can flow, with two outward arrows showing that it **emits light**
Switch ON / OFF	A closed or open gap in the wire
Wire	A straight line

A representation of a circuit using symbols is a **circuit diagram**. Drawing circuits as diagrams makes them quick to draw and easy to read. International bodies such as the **IEC, ANSI and IEEE** standardise symbols so people everywhere understand the same diagram.

3.4 Electrical Conductors and Insulators

Make a tester (Activity 3.11)
Connect a cell and lamp with two free wire ends. Touching the free ends together makes the lamp glow, so the tester works.

Test objects
Touch the free ends to both ends of each object (spoon, coin, cork, rubber, glass, key, pin, plastic scale, wooden block, aluminium foil, candle, needle, cardboard, paper, pencil lead). Make sure the wires do not touch each other.

Conductors (lamp glows)	Insulators (lamp does not glow)
Metal key, spoon, coin, pin, needle, aluminium foil, pencil lead (graphite)	Wood, plastic scale, glass bangle, paper, wax candle, rubber eraser, cork, cardboard

DEFINITION

Conductors (good conductors) allow electric current to flow easily. **Insulators (poor conductors)** do not allow current to pass through them.

- Metals are conductors. **Silver, copper and gold** are the best conductors, but **copper** is used for wires because it is cheaper and plentiful.
- Plastic, rubber and ceramics are **insulators**; they cover wires, plug tops and switches to protect us from shocks.
- Wires, switch parts, plug connectors and sockets are **conductors**; their coverings are **insulators**: both are important.

IMPORTANT

Our body is a conductor. Current through the body can cause severe injury or death. Never touch switches or plugs with wet hands, never use electrical devices in wet places and never handle damaged wires or broken plugs.

NOTE

DC and AC: batteries give **Direct Current (DC)** and power small devices. Power plants supply **Alternating Current (AC)** to wall sockets, and it can run larger appliances.

Science and society: cells come in different forms: cylindrical cells for torches, clocks, remotes and toys; button cells for watches and hearing aids; rechargeable batteries for phones, laptops and electric vehicles.

Reasoning questions from the chapter

Likely question	Model answer
Why does a torch lamp glow in only one position of the switch?	In the ON position the circuit is complete and current flows through the lamp; in the OFF position there is an air gap.
How are two cells joined in a battery?	Positive terminal of one cell touches the negative terminal of the next.

Likely question	Model answer
Why does the LED not glow when its wires are interchanged?	Current passes through an LED in one direction only; the longer wire must connect to the positive terminal.
Will a lamp glow if the insulation is not stripped from the wire ends?	No: the insulating cover blocks the contact, so no complete circuit forms.
If one filament in a circuit is broken, will the other lamp glow?	In a single series loop, no, because the circuit is broken; if the lamps are on separate paths, the other still glows.
How can you find the terminals of a battery whose signs have faded?	Connect it to an LED: it glows when the longer wire is joined to the positive terminal; this identifies the terminals.
Why are wires made of copper and covered with plastic?	Copper is a good, cheap conductor; plastic is an insulator that protects against shock and short contact.
What is a circuit diagram?	A drawing of a circuit using standard symbols.
Give four reasons why a lamp may not glow when connected.	Fused filament, loose connection, dead cell, wrong terminals/LED reversed, open switch, broken wire.
State the direction of current in a circuit.	From the positive terminal to the negative terminal of the cell, through the circuit.
Differentiate DC and AC.	DC flows in one direction (batteries, small devices); AC (wall socket) is used for larger appliances.

Exploratory projects

- List the actions you could not do in your daily life if power failed for two days.
- Run a toy fan with a solar panel.
- Visit an electrical shop to list types of cells and their uses.
- Classify home objects as insulators only, conductors only, or both.

★ KEY TAKEAWAYS

- A **cell** has + (metal cap) and – (flat disc) terminals; two or more cells joined + to – form a **battery**.
- An **incandescent lamp** glows by a hot **filament**; an **LED** has no filament and works only when connected the correct way (**longer wire to +**).
- A lamp glows only in a **closed circuit**; current flows from **+ to –**.
- A **switch** closes (ON) or opens (OFF) the circuit.
- A **circuit diagram** uses standard symbols.
- **Conductors** (metals, copper best for wires) let current pass; **insulators** (plastic, rubber, glass, wood) do not.
- The human body conducts electricity: handle electricity with great care.

CHAPTER 03

The World of Metals and Non-metals

Yashwant and Anandi visit a village ironsmith, Sudarshan uncle, who heats iron in a coal furnace and hammers it into tawas, buckets, axes and tools. This raises questions about why metals can be beaten into shape, how they differ from other materials, and how they behave in air and water. The chapter classifies elements into metals and non-metals using their physical and chemical properties.

Big Questions

- Why can iron be hammered into an axe while coal crumbles?
- Why are cooking vessels metal but their handles wood or plastic?
- What exactly is rust and how can we stop it?
- Why are metal oxides basic and non-metal oxides acidic?

4.1 Properties of Materials

Items tested in the chapter's activities are copper, aluminium, an iron nail, coal, sulfur (gandhak) and wood. Their behaviour under hammering, sound, heat and electricity reveals the physical properties that separate **metals** from **non-metals**.

▸ 4.1.1 Malleability and lustre (Activity 4.1)

Material	Lustrous?	Hard?	On hammering
Copper	Yes	Hard	Flattens
Aluminium	Yes	Hard	Flattens
Iron nail	Yes	Hard	Flattens
Coal	No	Less hard	Breaks into pieces (brittle)
Sulfur	No	Soft	Breaks into pieces (brittle)
Wood	No	Less hard	Neither flattens nor breaks

DEFINITION

Metallic lustre is the shine of a metal's surface. **Malleability** is the property by which a material can be beaten into thin sheets. Materials that break into pieces on hammering are **brittle**.

- Examples of malleability: silver foil on sweets, aluminium foil for wrapping food.
- **Gold and silver** are the most malleable metals.
- Not every metal is hard or solid: **sodium and potassium** are so soft that they can be cut with a knife, and **mercury** is liquid at room temperature (used in thermometers).
- **Holistic lens:** Harappans used copper and gold but hardly any iron. Iron came into everyday use later; its strength made ploughs and tools much better, which advanced Indian civilisation.

▸ 4.1.2 Ductility

> DEFINITION
> **Ductility** is the property by which a material can be drawn into wires. It is mainly a property of metals.

- Uses: electrical wires (copper, aluminium), ornaments, strings of the veena, sitar, violin and guitar, tea strainers.
- **Gold is so ductile that 1 g can be drawn into a wire 2 km long.**
- Coal and sulfur are not ductile.
- Steel (a mixture of iron, a metal, and carbon, a non-metal) wire ropes carry heavy loads in suspension bridges and cranes.

▸ 4.1.3 Sonority (Activity 4.2)

Drop a metal spoon, coin, piece of coal and wooden block from a height. Metals make a **ringing sound**; coal and wood make a dull sound. The ability of metals to produce a ringing sound is **sonority**; metals are **sonorous**. School bells and ghungroos ring because of this.

▸ 4.1.4 Conduction of heat (Activity 4.3)

> **Set up**
> Fill a glass tumbler with hot water (be careful).

> **Immerse**
> Put a metal spoon and a wooden spoon of similar size at the same time and leave for a few minutes.

> **Touch**
> Carefully touch the upper ends. The metal spoon is hotter.

Conclusion: heat passes easily along the metal spoon, but poorly along the wooden spoon. The transfer of heat from one point to another through a material is **conduction**. **Metals are good conductors of heat**; wood is a poor conductor. So cooking vessels are metal, but handles are wood or other poor conductors.

▸ 4.1.5 Conduction of electricity (Activity 4.4)

Make the tester circuit from the previous chapter and test aluminium foil, iron nail, sulfur, copper wire, coal, dry wood, stone, rubber eraser and nylon rope. The bulb glows only with **aluminium, iron and copper**, so all materials that conduct electricity are metals.

- **Good conductors of electricity** let current pass easily; **poor conductors** do not.
- Screwdriver handles are plastic and electricians wear rubber gloves and shoes because plastic and rubber are poor conductors and protect from shocks.

> IMPORTANT
> **Metals are generally hard, lustrous, malleable, ductile, sonorous and good conductors of heat and electricity.** Exceptions: sodium and potassium are soft; mercury is a liquid.

4.2 Effect of Air and Water on Metals: Iron

Iron objects left in the open develop a brown deposit. Activity 4.5 finds out which conditions are needed.

Bottle	Condition	Air present?	Water present?	Rust?
A	Nail with silica gel (dry air), tightly capped	Yes (dry)	No	No

Bottle	Condition	Air present?	Water present?	Rust?
B	Nail in boiled, cooled water with oil layer on top, tightly capped	No	Yes	No
C	Nail partly dipped in water, open	Yes	Yes	Yes

- Silica gel keeps air dry; boiled water has no dissolved gases; oil prevents air from dissolving in water.
- Observe for 8–10 days. Rust forms only in C.

DEFINITION

Rust is the brown deposit on iron formed in the presence of **both air (oxygen) and water (moist air)**; the process is **rusting**. **Corrosion** is the gradual damage of metal surfaces by air, water or other substances.

- Other examples of corrosion: green coating on copper, black coating on silver.
- Rusting costs a huge amount in replacing or repairing iron structures.
- **Prevention:** painting, oiling, greasing, and **galvanisation** (coating iron with zinc).
- **Iron Pillar of Delhi:** made in the time of Chandragupta II over 1600 years ago, about 8 m high, weighs over 6000 kg, and has barely rusted, showing India's advanced metal technology.

4.3 Effect of Air and Water on Other Metals

Burn magnesium (teacher demonstration)
Clean a 3–4 cm magnesium ribbon with sandpaper, hold it with tongs and ignite it. Wear protective goggles and keep distance.

Observe
It burns with a **dazzling white flame** to leave a white powder, **magnesium oxide**, formed from magnesium and oxygen of air.

Test the product
Dissolve the powder in a few drops of warm water and test with litmus. It turns **red litmus blue**, so it is **basic**.

IMPORTANT

Oxides of metals are generally basic in nature.

- **Sodium** reacts vigorously with oxygen and water giving a lot of heat, so it is stored in **kerosene** to keep out air and moisture.

4.4 Substances that Behave Differently from Metals

Burn sulfur (teacher demonstration, ventilated area)
Burn powdered sulfur in a deflagrating spoon and lower it into a covered gas jar. Burning sulfur produces harmful gas.

Dissolve the gas
After 3–4 minutes remove the spoon, add a little water and shake.

Test
The solution turns **blue litmus red**, so it is **acidic**. Sulfur + oxygen → sulfur dioxide; sulfur dioxide + water → sulfurous acid.

Sulfur with water
Sulfur powder in water shows no reaction.

- Sulfur and phosphorus are **non-metals**. **Phosphorus** catches fire in air, so it is stored **under water**.
- Non-metals are usually soft and dull, neither malleable nor ductile, not sonorous, and poor conductors of heat and electricity. Examples: oxygen, hydrogen, nitrogen, carbon, sulfur, phosphorus.
- **Oxides of non-metals are acidic.** Non-metals generally do not react with water.
- Plastic, glass, wood, rubber and paper are **not** metals or non-metals because they are not elements.
- An **element** is a substance that cannot be broken down into simpler substances; **118 elements** are known, some artificial. Metals and non-metals are sub-categories of elements.

Property	Metals	Non-metals
Lustre	Lustrous	Generally dull
Hardness	Generally hard (exceptions: Na, K)	Generally soft (carbon as diamond is hard)
Malleable / ductile	Yes	No; brittle if solid
Sonority	Sonorous	Not sonorous
Heat and electricity	Good conductors	Poor conductors
Oxide	Basic (turns red litmus blue)	Acidic (turns blue litmus red)
With water	Some react (Na vigorously)	Generally do not react
Examples	Iron, copper, aluminium, gold, sodium	Sulfur, carbon, oxygen, nitrogen, phosphorus

4.5 Are Non-metals Essential in Everyday Life?

Non-metal	Importance
Oxygen	Needed for breathing and survival
Carbon	Building block of life; in proteins, fats, carbohydrates
Nitrogen	Making fertilisers; nutrient for plant growth
Chlorine	Water purification
Iodine	Solution used as antiseptic on wounds

Science and Society: alloys (mixtures of two or more metals, or a metal and a non-metal) are used in utensils, tools and industry. Special metals are used in atomic energy (zirconium) and aerospace (titanium). In India iron and aluminium are widely **recycled** to cut waste.

Likely exam questions

Likely question	Model answer
What are malleability and ductility?	Malleability is the ability to be beaten into thin sheets; ductility is the ability to be drawn into wires.
Name the two most malleable metals.	Gold and silver.

Likely question	Model answer
Name a metal that is liquid at room temperature and metals that can be cut with a knife.	Mercury is liquid; sodium and potassium are soft.
Why are cooking vessels metal but handles wood/plastic?	Metals conduct heat well, so food heats up; wood/plastic are poor conductors, so handles stay cool.
What is the lesson of the rusting experiment?	Rust forms only when iron is in contact with both air and water.
How can rusting be prevented?	Painting, oiling, greasing, galvanisation (zinc coating).
Why is sodium kept in kerosene and phosphorus in water?	Sodium reacts vigorously with air/moisture; phosphorus catches fire in air.
What happens when magnesium burns and what is the product's nature?	Burns with white flame to form magnesium oxide, which is basic (turns red litmus blue).
What happens when sulfur burns and the product is dissolved in water?	Sulfur dioxide forms and gives sulfurous acid, which is acidic (turns blue litmus red).
Why is sulfur unsuitable for galvanisation?	A protective coating needs a metal like zinc; sulfur is a brittle non-metal, not a coating metal.
Why does an ironsmith heat iron before shaping?	Hot iron becomes soft and malleable, so it can be hammered easily into tools.
Which non-metals are essential in daily life and why?	Oxygen (respiration), carbon (life), nitrogen (fertilisers), chlorine (purification), iodine (antiseptic).

Exploratory projects

- Find the states of Dhokra, Bidriware, Pembarthi and Kamrupi metal art and make a collage.
- Mark the states where iron, gold and aluminium are found on a map of India.
- List metals and non-metals used in a smartphone.
- Debate: should metals for comfort and luxury be used more or less?

★ KEY TAKEAWAYS

- Metals are lustrous, hard, **malleable, ductile, sonorous** and good conductors of **heat and electricity**.
- Non-metals are generally dull, brittle or soft and poor conductors.
- Iron **rusts** only in the presence of **air and water**; corrosion is damage of metal surfaces; prevent with painting, oiling, greasing or galvanisation.
- **Metal oxides are basic; non-metal oxides are acidic.**
- Sodium is stored in kerosene, phosphorus under water.
- **Elements** are the basic substances (118 known) divided into metals and non-metals; plastic, wood, glass are not elements.
- Non-metals like oxygen, carbon, nitrogen, chlorine and iodine are essential for life.

CHAPTER 04

Changes Around Us: Physical and Chemical

Ice melts, buds bloom, bananas turn brown and cold water warms up: changes are everywhere. This chapter sorts changes into physical and chemical, studies rusting and combustion, looks at the fire triangle, shows that one process can involve both kinds of change, and ends with reversible and irreversible changes, desirable and undesirable changes, and slow natural changes like weathering and erosion.

Big Questions

- How can we tell whether a new substance has formed?
- What three things are needed for fire?
- Is burning a candle a physical or a chemical change?
- How are rocks changed into soil?

5.0 Changes we observe

Activity 5.1 lists changes such as melting ice, chopping vegetables, boiling water, making popcorn, cutting paper, adding beetroot extract to water, burning wood, drying clothes, making dough balls and rolling chapatis. Each causes a change in size, shape, smell, colour, state or another property. We notice changes through our senses; the aim is to arrange them into categories.

5.1 A Substance May Change in Appearance but Remain the Same

Activity 5.2: folding paper into shapes, inflating and deflating a balloon, and crushing chalk into powder. In each case the material (paper, rubber, chalk) stays the same even though shape or size changes. Water too changes between solid, liquid and gas but remains water.

DEFINITION

A **physical change** is a change in physical properties (shape, size, state, etc.) in which **no new substance is formed**.

5.2 A Substance May Change in Appearance and Not Remain the Same

Prepare

Take two tumblers: A with a little tap water, B with lime water.

Blow

Gently exhale through a straw into each (never suck the liquid).

Observe

A: only bubbles, no change. B: lime water turns **milky**; on standing a white solid settles. Something new has formed.

Exhaled **carbon dioxide** reacts with lime water (calcium hydroxide) to form **calcium carbonate**, a white solid insoluble in water, plus water.

FORMULA

Calcium hydroxide (lime water) + Carbon dioxide → Calcium carbonate (insoluble) + Water

DEFINITION

A **chemical change** is one in which **one or more new substances are formed**. The process is a **chemical reaction**, written as a **chemical equation**. Milkiness of lime water is the **test for carbon dioxide**.

▸ Vinegar and baking soda (Activity 5.4)

- Add a pinch of baking soda (sodium hydrogen carbonate) to vinegar or lemon juice: fizzing and gas bubbles.
- Pass the gas through fresh lime water: it turns milky, so the gas is **carbon dioxide**.
- Vinegar + Baking soda → Carbon dioxide + Other substances: a chemical change.
- Baking soda in plain water gives no bubbles.

5.3 Other Processes Involving Chemical Changes

▸ 5.3.1 Rusting

Rust, a brown substance (iron oxide), is formed when iron meets air and water, so rusting is a chemical change.

▸ 5.3.2 Combustion

Burning magnesium ribbon forms magnesium oxide with heat and light, so it is a chemical change.

FORMULA

Magnesium + Oxygen → Magnesium oxide (white powder) + Heat + Light

DEFINITION

Combustion is a chemical reaction in which a substance reacts with oxygen to give heat and/or light. Substances that burn are **combustible** (wood, paper, cotton, kerosene).

▸ Is oxygen needed? (Activity 5.5)

Two identical candles are lit; one is covered by a glass tumbler. The covered one goes out after a while because air (oxygen) is cut off. Lime water added in the tumbler turns milky: carbon dioxide formed from wax carbon and oxygen of air. **Oxygen is essential for combustion.**

IMPORTANT

Putting out clothes on fire: wrap the person in a blanket or cloth to cut off air. Never use synthetic cloth: it melts and sticks to skin.

- **Bioluminescence:** fireflies emit light (without heat) by a chemical change.

▸ Is air enough? Ignition temperature (Activity 5.6)

A lit match sets paper on fire instantly. Focusing sunlight with a magnifying glass also makes paper smoke and burn, so fire is not needed to start burning; heat is. The paper warms until it reaches its **ignition temperature**.

DEFINITION

Ignition temperature is the minimum temperature at which a substance catches fire.

Requirement for combustion (fire triangle)	Meaning
Combustible substance (fuel)	Something that can burn
Oxygen	Supplied by air
Heat	Raises the fuel to its ignition temperature

Removing any side of the triangle puts out the fire.

5.4 Can Physical and Chemical Changes Occur in the Same Process?

In a burning candle the wax **melts** (physical), rises through the wick, **evaporates** (physical), and the **wax vapour burns** (chemical). Solidifying of dripping wax is physical. So the burning of a candle involves both kinds of change.

Know a scientist: Michael Faraday gave the nineteenth-century lectures *Chemical History of a Candle*, using a candle to teach melting, vapourisation and combustion.

5.5 Are Changes Permanent?

Reversible (original can be obtained)	Irreversible (cannot be brought back)
Melting ice (refreezes), boiling water (steam condenses), dissolving sugar in water, folding a mat, twisting string	Chopping vegetables, making popcorn, cooking, idli from batter, grinding wheat, ripening fruit, curdling milk, burning, rusting, formation of soil

> **NOTE**
> Physical changes are often reversible, but not always (chopping, grinding). Chemical changes are generally irreversible.

5.6 Are All Changes Desirable?

- **Desirable:** milk to curd, fruit ripening, cooking food, cutting fruit.
- **Undesirable:** rusting of iron, decay of stored food. A change can be good in one setting: decomposition of food waste makes **compost**.
- **Long-term harm from human activity:** burning more fuel in vehicles raises carbon dioxide in the air; drying paint releases substances that pollute the air.

5.7 Some Slow Natural Changes

▸ 5.7.1 Weathering of rocks

- Heaps of sand, soil and stones at the base of mountains are **sediments**.
- **Physical causes:** temperature changes, growing tree roots, water freezing in cracks.
- **Chemical causes:** water or chemicals in water react with rocks, e.g. black iron-containing basalt gets a red layer of **iron oxide**.
- Physical and chemical breaking of rocks together is **weathering**, which finally forms **soil**.

▸ 5.7.2 Erosion

- **Erosion** is the breaking down and movement of rocks, pebbles, soil and sediments by wind and flowing water.

- River pebbles are smooth because of constant erosion; landslides are an example (physical change).
- When water or wind slows (lake, ocean), material settles, and the sediments harden over time into **new rocks**.
- These changes take thousands of years and are not reversible.

Feature	Physical change	Chemical change
New substance	No	Yes
What changes	Shape, size, state, appearance	Composition and properties
Reversibility	Often reversible	Usually irreversible
Energy	Usually little	Often heat, light, gas, colour or smell
Examples	Melting, tearing, folding, dissolving sugar	Rusting, burning, curdling, ripening, baking soda + vinegar

Likely exam questions

Likely question	Model answer
Define physical and chemical change with examples.	Physical: no new substance (melting ice). Chemical: new substance formed (rusting).
How do you test for carbon dioxide?	Pass it through fresh lime water; it turns milky.
What is formed on exhaling into lime water?	White calcium carbonate, so the lime water turns milky.
What is combustion?	Reaction of a substance with oxygen producing heat and/or light.
What are the three requirements for combustion?	Fuel, oxygen and heat to reach ignition temperature (fire triangle).
Define ignition temperature.	The lowest temperature at which a substance catches fire.
Why does a covered candle go out?	Air (oxygen) is cut off.
Is the burning of a candle physical or chemical?	Both: melting and evaporation of wax are physical; burning of vapour is chemical.
Is curdling of milk physical or chemical?	Chemical; a new substance (curd) with different properties forms and cannot be reversed.
Are ice to water and water to steam chemical?	No, both are physical changes of state; no new substance is formed.
Differentiate weathering and erosion.	Weathering is physical/chemical breaking of rocks to form soil; erosion is breaking down and carrying away of material by wind or water.
Why is rusting a chemical change?	A new brown substance, iron oxide, forms.

Exploratory projects

- Observe the kitchen and list reversible changes.

- Yeast project: sugar solution + yeast in a bottle with a balloon on top gives a gas that turns lime water milky (carbon dioxide).
- Lemon-juice invisible ink: writing appears brown on warming (a chemical change).
- Discuss steps to reduce landslides and rock erosion.

★ KEY TAKEAWAYS

- **Physical change:** no new substance; **chemical change:** new substance(s) formed through a chemical reaction.
- Lime water turning milky tests for **carbon dioxide**.
- **Combustion** needs fuel, oxygen and heat (reaching **ignition temperature**) and gives heat and/or light.
- Rusting, burning, cooking, curdling are chemical changes.
- A candle shows physical and chemical changes together.
- Some changes are reversible, some not; some desirable, some not.
- **Weathering** forms soil from rocks; **erosion** moves material by wind and water over thousands of years.

CHAPTER 05

Adolescence: A Stage of Growth and Change

Plants and animals must grow and mature before they can reproduce. Humans pass through infancy, childhood, adolescence, adulthood and old age, each at their own pace. This chapter studies adolescence (roughly 10 to 19 years): the visible and internal body changes, emotional and behavioural changes, hormones, and how to make this stage healthy and joyful.

Big Questions

- Why do voices change and pimples appear?
- What is puberty and what are secondary sexual characteristics?
- Why must we say NO to harmful substances?
- Which chemicals control these changes?

6.0 Life stages

- Human life stages: **infancy, childhood, adolescence, adulthood, old age**; every person goes through them at their own speed.
- Until about 10–12 years, changes are mostly in height and weight. After this, other noticeable changes begin, marking the start of adolescence.
- A body must mature before it can reproduce, as seeds appear only on a mature plant.

DEFINITION

Adolescence is the stage of rapid growth and development between childhood and adulthood, generally from about **10 to 19 years** of age. The body prepares for adulthood with physical, emotional and behavioural changes.

6.1 Growing with Age: The Teenage Years

Activity 6.1 asks students to anonymously write on slips the changes noticed from Grade 5 to 8, then tabulate the commonest (height, weight and strength, appearance and so on). The commonest observable changes are given below.

Change	In boys	In girls
Height	Marked increase during adolescence	Marked increase during adolescence
Body structure, weight and strength	Broader shoulders and wider chest; weight gain	Weight gain; development of breasts
Voice	Growth of the **voice box (larynx)** makes voice hoarse; seen as bump called **Adam's apple** (not noticeable in all)	Voice box also grows but less; only slight change in voice
Body hair	Hair in armpits and pubic region; facial hair (moustache, beard); sometimes chest and back hair	Hair in armpits and pubic region
Skin	Acne and pimples	Acne and pimples

- **Acne** is a skin condition; pimples are its manifestation. Increased **oily secretions** clog skin pores and cause infection.
- **Variations are normal:** the time, nature and extent of changes differ from person to person, and so does the duration of adolescence.

DEFINITION

Secondary sexual characteristics are features that distinguish males from females but are not directly involved in reproduction (e.g. voice change, moustache and beard, breast development). They show the body is preparing for adulthood and mark the onset of **puberty**. **Puberty** is the stage when the body undergoes external and internal changes to become an adult capable of reproduction.

6.2 Changes that Indicate Reproductive Capability

Along with visible changes, the organs of reproduction mature gradually inside the body. In girls, an important internal change is the **menstrual cycle**.

Fact	Detail
Menstrual cycle	Recurs generally every **28–30 days** (healthy cycles may vary from 21–35 days); popularly called 'the period'
Menstruation	The phase when blood discharge occurs; lasts about **3 to 7 days**
Discomfort	Some girls feel lower-abdominal pain during these days
Starts at	Puberty
Stops naturally at	About **45–55 years**, ending reproductive capability
Significance	A natural process and a sign of good reproductive health

IMPORTANT

Myths about menstruation (e.g. isolating menstruating girls) have no scientific basis. A scientific outlook removes fear, shame and guilt and supports women's reproductive health.

6.3 Emotional and Behavioural Changes

Adolescents have stronger emotions than children. Understanding how emotions affect behaviour helps us choose well and respond thoughtfully.

Emotional change	Possible effect on behaviour	Positive ways to grow
Mood swings	Feeling very energetic and happy one day, low on another	Music, dance, sports and varied activities
Strong emotions	Deep interest in new areas; urge to explore oneself	Self-exploration and activities leading to creativity and innovation
Increased sensitivity	Concern for the needy and disadvantaged	Compassion; social work and social initiatives

6.4 Making Adolescence a Joyful Experience

▸ 6.4.1 Meeting nutritional needs

Rapid growth demands a balanced diet with carbohydrates, proteins, fats, vitamins and minerals.

Food sources	Nutrient	Function
Milk, millets, curd, cheese, paneer	Calcium, proteins, fats	Optimal bone growth
Pulses, eggs, nuts and similar protein foods	Proteins	Growth, strength, energy
Spinach, kidney beans, raisins, figs	Iron	Formation of blood

- Adolescents, especially girls, can suffer blood-related problems (anaemia) from **iron or vitamin B12 deficiency**.
- **Vitamin B12** cannot be made by the body; it must come from food. **Dorothy Hodgkin** studied its structure and in 1964 became the third woman to win the Nobel Prize in Chemistry.
- Some neck swellings in adolescents can be linked to iodine, which is why an iodine-rich diet may be advised; a doctor decides the reason in each case.

▸ 6.4.2 Personal hygiene

- Keep the body clean, especially the armpits and pubic region, to prevent infections.
- During menstruation use **sanitary pads or reusable cloth pads** and change regularly.
- Dispose of used pads properly: wrap in newspaper and put in a dustbin. **Biodegradable pads** are eco-friendly.
- Society should reduce stigma and provide facilities in schools and public places.

Government initiative	Aim
Menstrual Hygiene Scheme (MHS)	Free or low-cost sanitary pads and awareness for adolescent girls in rural areas
Rashtriya Kishor Swasthya Karyakram (RKSK)	Overall adolescent health including menstrual health; peer education
Suvidha Sanitary Napkin Initiative	Affordable biodegradable pads through Jan Aushadhi Kendras
State schemes	e.g. Shuchi Scheme (Karnataka); free napkin schemes in Tamil Nadu and Odisha

▸ 6.4.3 Physical activities

Regular exercise, games and sports keep body and mind fit, build **stamina** and improve mood.

▸ 6.4.4 Balanced social life

- Be polite and respectful in person and online; adolescents may feel attracted to peers and copy their behaviour.
- Use online platforms responsibly; seek guidance from elders and teachers.
- **Cyberbullying** is using phones, computers or online platforms to harass others by false messages, rumours or sharing personal information without consent. If bullied, do not be scared: seek help from parents and teachers. Be cautious about uploading pictures and sharing personal information with strangers.

Social media dos	Social media don'ts
Be respectful and kind; think before posting; protect privacy	Do not share personal photos or information with strangers or virtual friends; do not spread rumours

6.4.5 Avoiding harmful substances: learn to say NO

- Peers may pressure you to try tobacco, gutka, cigarettes, beedi, alcohol or illegal drugs. They harm physical and mental health and are **addictive** (a strong urge to take them again and again).
- Regular use is **substance abuse**. Effects include breathing difficulty, memory loss and lung damage.
- Say NO the first time and every time. Seek help from family, parents, teachers, counsellors and doctors.
- **Nasha Mukt Bharat Abhiyaan** (Ministry of Social Justice and Empowerment) spreads awareness, with special focus on early-age children. **National De-addiction Helpline: 14446.**

6.5 The 'Why' of Adolescence: Hormones

DEFINITION

Hormones are chemicals produced in different parts of the body and released at the right time on signals from the brain. They regulate growth and development. Menstruation and puberty changes are mainly due to hormones; some hormones also influence mood and behaviour.

Likely exam questions

Likely question	Model answer
Define adolescence and give its age range.	The stage between childhood and adulthood, roughly 10 to 19 years.
What are secondary sexual characteristics? Give examples.	Features distinguishing males from females but not directly used in reproduction: voice change, moustache and beard in boys, breast development in girls.
What is puberty?	The stage when the body changes externally and internally to become capable of reproduction.
Why does a boy's voice become hoarse?	The voice box grows; the bump is the Adam's apple.
What causes pimples?	Increased oily secretions clog skin pores and lead to infection (acne).
What is menstruation and how often does the cycle recur?	Discharge of blood from the body in girls during a cycle that recurs every 28–30 days; it lasts 3–7 days and ends by 45–55 years.
Is Shalu right that adolescence brings only physical changes?	No, it also brings emotional, behavioural and reproductive changes.
Give positive outlets for mood swings.	Music, dance, sports, creative activities, social service.
Why is a balanced diet important in adolescence?	Rapid growth needs proteins, carbohydrates, fats, vitamins and minerals (calcium for bones, iron for blood).
How should used sanitary pads be disposed of?	Wrap in paper and put in a dustbin; biodegradable pads are better.

Likely question	Model answer
What is cyberbullying and what should you do?	Harassing others online; do not feel helpless, seek help from parents and teachers.
Why should one avoid trying harmful substances 'just once'?	They are addictive, harm body and mind; one try may lead to regular use (substance abuse).
What are hormones?	Chemicals made in the body that control growth, puberty and mood.

Exploratory projects

- Find personalities and organisations working for youth mental health in your locality; list five interview questions.
- Role play 'Child marriage: A social evil' on its harm to children's health.
- Organise a yoga camp for International Yoga Day (21 June).

★ KEY TAKEAWAYS

- Adolescence (about **10–19 years**) is the bridge between childhood and adulthood.
- Changes: height, weight, voice, body hair, acne, **secondary sexual characteristics**; **puberty** brings reproductive capability.
- Menstrual cycle recurs about every **28–30 days**; menstruation lasts 3–7 days; myths are baseless.
- Emotional changes include mood swings and strong feelings; channel them into positive activities.
- Healthy habits: balanced diet, hygiene, exercise, responsible social life, say **NO** to addictive substances.
- **Hormones** control these changes.
- Every person develops at their own pace; variations are normal.

CHAPTER 06

Heat Transfer in Nature

Pema and Palden sit around a fireplace in Gangtok and wonder why Kerala is warmer, why cooking pans are metal and why smoke rises. The chapter explains the three ways heat travels (conduction, convection and radiation), applies them to clothing, houses and sea/land breezes, and then links the Sun's heat to the water cycle, infiltration, groundwater and aquifers.

Big Questions

- Why do we use metal pans but clay cups?
- Why does smoke rise?
- Why does the wind blow from sea to land in the day?
- How does rainwater become groundwater?

7.0 Why places differ in temperature

The Sun is the main source of heat and light for the Earth. Places near the **equator** are generally hot, and coastal places like Kerala are warm and humid, while hilly places like Sikkim are cold in winter. Cooking utensils are metal because metals are good conductors of heat.

7.1 Conduction of Heat

Set up (Activity 7.1)
Attach four pins with wax at about 2 cm gaps along a 15 cm aluminium or iron strip, clamp it on a stand and label the pins I to IV (I nearest the flame).

Heat
Heat the free end with a candle or spirit lamp, under adult supervision.

Observe
Pin I falls first, then II, III and IV. Heat travels along the strip and melts the wax one pin at a time.

DEFINITION

Conduction is the transfer of heat from the hotter part of an object to the colder part. Each particle passes heat to its neighbour but **the particles themselves do not move from their positions**. Materials that conduct heat easily are **good conductors**; those that do not are **poor conductors (insulators)**.

- Conduction is the **main mode of heat transfer in solids**.
- With a wood or glass strip the pins would not fall because they are poor conductors.
- Metals are used for cooking vessels; **clay and porcelain** cups are used for tea and coffee, which stay hot longer because they conduct poorly.

Good conductors of heat	Poor conductors (insulators) of heat
Steel, iron, aluminium, copper and other metals	Wood, glass, clay, porcelain, plastic, air

▸ Applications of poor conductors

- **Woollen clothes** trap air in their pores; air is a poor conductor, so heat from the body escapes slowly and we feel warm.

- **Two thin blankets** keep us warmer than one thick one because of the air layer trapped between them.
- **Mori block, Uttarkashi (Himalayas):** walls of two wooden layers filled with cow dung and mud; wood and mud are poor conductors, so they prevent heat loss in winter.
- **Hollow bricks:** trapped air keeps houses warm in winter and cool in summer.

7.2 Convection

Paper cups (Activity 7.2)
Hang two identical inverted paper cups from the ends of a balanced stick.

Heat
Place a burning candle below one cup.

Observe
That cup rises. The warmed air expands, occupies more space, becomes lighter and rises.

- A partly inflated balloon kept in the Sun grows larger: air **expands on heating**.
- Smoke from burning incense rises as it is a mixture of hot gases and solid particles, warmer than surrounding air. A **smoke detector** is therefore best placed on the **ceiling**.

Liquids (Activity 7.3)
Half-fill a beaker with water and put a grain of potassium permanganate at the centre of the base with a straw.

Heat from below
Place a candle under the centre of the beaker.

Observe
A coloured streak rises in the middle and comes down at the sides. Hot water at the bottom expands, becomes lighter and rises; cooler, heavier water from the sides sinks to replace it. The cycle continues until all the water is hot.

DEFINITION

Convection is the transfer of heat by the **actual movement of particles** of liquids and gases from one place to another. Water and air are heated by convection.

▸ 7.2.1 Land and sea breeze (Activity 7.4)

Soil and water of the same amount are put in sunlight with a thermometer in each. After 20 minutes the soil is much hotter than the water; indoors the soil also cools faster. So **land heats and cools faster than water**.

	Sea breeze	Land breeze
Time	Day	Night
Which is warmer	Land (heats faster)	Sea (cools slower)
Air movement	Warm air above land rises; cooler air moves from **sea to land**	Warm air above sea rises; cooler air moves from **land to sea**
Effect	Relief from heat; houses on coasts have windows facing the sea	Wind direction reverses at night

Large water bodies also keep nearby areas from reaching extreme temperatures, as water heats and cools slowly.

7.3 Radiation

DEFINITION

Radiation is the transfer of heat directly from a hot object without needing any medium. The heat of the Sun reaches the Earth, and the warmth of a fire reaches us, by radiation. **All objects radiate heat**; a hot utensil cools by radiating heat to its surroundings.

- **Light-coloured clothes** reflect most heat, so they are comfortable in summer.
- **Dark-coloured clothes** absorb more heat, so they suit winter.

Mode	Medium needed?	Particle movement	Main occurrence	Example
Conduction	Yes	Particles vibrate in place, passing heat on	Solids	Spoon in hot tea, metal strip
Convection	Yes	Particles actually move	Liquids and gases	Boiling water, land/sea breeze, rising smoke
Radiation	No	No particles involved	Even through vacuum/space	Sunlight, heat from a fire

Example of all three: in heating water in a pan, heat goes from the flame to the pan by conduction, from the pan to the water by convection, and the warmth felt around the flame and pan is by radiation. The Himalayan **bukhari** (iron stove with chimney, flat top for cooking) uses all three.

7.4 Water Cycle

The Sun's heat drives evaporation of water from clothes, oceans, rivers and lakes. Water exists as a liquid (oceans, rivers), solid (snow, glaciers, ice sheets) and gas (water vapour).

Evaporation
Sun heats oceans, rivers and lakes; water becomes vapour. Plants also lose water vapour by **transpiration**.

Condensation
Rising vapour cools and condenses into clouds.

Precipitation
Water falls as rain, snow or hail.

Return
Water flows into rivers, lakes and oceans or seeps into the ground; snow in mountains melts in summer and is replenished in winter.

DEFINITION

The **water cycle** is the continuous movement of water: upward as vapour and downward by precipitation, through soil, rocks and plants, finally back to water bodies. It redistributes and replenishes water and conserves the total water on Earth.

Know a scientist: Varahamihira (6th century CE, Ujjain) wrote *Brihatsamhita* and predicted seasonal rainfall using clouds, winds, and positions of stars and the moon.

▸ 7.4.1 Seepage of water beneath the Earth (Activity 7.5)

Three bottles filled with clay, sand and gravel are each given 200 mL water. **Water passes fastest through gravel, slower through sand and slowest through clay**, because spaces between gravel particles are wider.

DEFINITION

Infiltration is the seeping of surface water through soil and rocks. It is easier when spaces are wide, open and interconnected. Stored water in pore spaces is **groundwater**. The underground layers of sediments and rocks that hold it are **aquifers**. Wells and bore wells are dug into aquifers (a few metres to hundreds of metres deep).

- Groundwater is not unlimited: heavy extraction for a growing population and concrete surfaces in cities that reduce infiltration deplete it.
- Remedies: **rainwater harvesting** and **recharge pits**.
- **Ice stupa (Ladakh):** in winter, stream water is piped and sprayed in cold air to freeze into a cone-shaped ice structure, which melts slowly in spring to give water for farming through summer.

Likely exam questions

Likely question	Model answer
Name the three modes of heat transfer.	Conduction, convection and radiation.
Define conduction. In which state of matter is it the main mode?	Transfer of heat from hotter to colder part without particles moving; solids.
Why are cooking vessels made of metal but handles of wood/plastic?	Metals conduct heat well; wood/plastic are poor conductors.
Why do two thin blankets keep us warmer than one thick one?	Air trapped between the layers is a poor conductor and reduces heat loss.
Define convection with an example.	Heat transfer by actual movement of particles of liquids and gases; water in a pan, sea breeze.
Why does hot air rise?	On heating it expands, becomes lighter and rises.
Why should a smoke detector be on the ceiling?	Smoke is hot and rises.
Explain sea breeze and land breeze.	In the day land is warmer, so air flows from sea to land; at night sea is warmer, so air flows from land to sea.
Why does heat from the Sun reach us?	By radiation, which needs no medium.
Why do we wear light clothes in summer and dark in winter?	Light colours reflect heat; dark colours absorb it.
What is the water cycle?	Continuous movement of water by evaporation, condensation and precipitation, returning to water bodies and the ground.
What are infiltration, groundwater and aquifer?	Seeping of water into the ground; water stored in pore spaces; underground layers storing water.
Why do hollow bricks help in houses?	Trapped air is a poor conductor, so houses stay cool in summer and warm in winter.
Where do ice cubes in a dish get heat to melt?	From the surroundings (air, dish) by conduction, convection and radiation.

Exploratory projects

- Wrap a thin paper strip around a metal rod and try to burn it while rotating: the paper does not burn because the metal conducts away the heat.
- Hang a spiral-cut paper above a candle: it spins due to rising hot air (convection).
- Visit a water-harvesting site or recharge pit and prepare an illustrated report.

★ KEY TAKEAWAYS

- Heat moves by **conduction** (solids, particles stay), **convection** (liquids/gases, particles move) and **radiation** (no medium).
- Metals are good conductors; wood, air, glass, clay are poor conductors, used for insulation.
- Warm fluid rises and cool fluid sinks, giving **convection currents**, **sea breeze** (day) and **land breeze** (night).
- Land heats and cools faster than water.
- The Sun's radiation drives the **water cycle**: evaporation, condensation, precipitation.
- **Infiltration** recharges **groundwater** stored in **aquifers**; conserve it by rainwater harvesting and recharge pits.